



MAPUA MALAYAN
DIGITAL COLLEGE

FinMark

Sales Forecasting & Predictive Analytics

Terminal Assessment — Machine Learning & Predictive Analytics

Paran, Ralph Matthew

BSIT-S3101 | MO-IT134 | March 24, 2026

01

Problem Statement

02

Dataset & Preparation

03

Feature Engineering

04

Model Development

05

Results & Forecast

06

Recommendations

01 — Problem Statement & Objectives

Introduction



MAPÚA MALAYAN
DIGITAL COLLEGE

What problem are we solving?

The Problem

FinMark Corporation needs a data-driven way to anticipate sales performance. Without accurate forecasting, the company cannot:

- Allocate resources and budget effectively
- Identify when sales are declining early enough to act
- Understand which products and seasons drive the most revenue
- Make informed decisions about customer acquisition strategy

Our Objectives

Predict whether a customer will make a purchase (Milestone 1)

Forecast company sales for the next 6 months (Milestone 2)

Engineer meaningful features that improve model performance

Visualize trends and patterns to support business decisions

Recommend actionable strategies based on the model findings

02 — Dataset Description

Data Preparation



MAPÚA MALAYAN
DIGITAL COLLEGE

Three datasets, one complete picture of FinMark's business

customers_data.csv

100 rows · 4 columns

Company profiles — IDs, names, profit, address

Key: **Company_ID**

products_data.csv

20 rows · 3 columns

FinMark software catalog — IDs, names, prices

Key: **Product_ID**

transactions_data.csv

10,000 rows · 8 columns

Full transaction log — dates, quantities, costs

Key: **Company_ID + Product_ID**

Data Cleaning & Preprocessing

- Dropped rows with missing ID columns (cannot identify or join)
- Filled missing numeric values (profit, price, quantity) with column median
- Stripped currency symbols from Product_Price (e.g. '?140,000' → 140000)
- Parsed 10+ mixed date formats using pandas format='mixed'
- Removed auto-generated unnamed index column from transactions
- Post-merge: filled unmatched rows with median (numeric) or 'Unknown' (text)

03 — Feature Engineering



MAPUA MALAYAN
DIGITAL COLLEGE

Data Preparation

8 new features created from existing columns

Year

Source: Transaction_Date

Detects multi-year growth or decline

Month

Source: Transaction_Date

Captures seasonal sales patterns

Quarter

Source: Transaction_Date

Reflects quarterly B2B budget cycles

Month_Number

Source: Year + Month

Sequential index (1-60) — main model input

Revenue_Per_Unit

Source: Total_Cost ÷ Quantity

Detects pricing or product mix shifts

Is_High_Value

Source: Total_Cost > median

Flags premium transaction patterns

Product_Name

Source: Join from products_data

Enables product-level revenue analysis

Rolling_3M_Avg

Source: 3-month rolling mean

Smooths noise to reveal true trends ★
NEW

03 — Exploratory Data Analysis



MAPÚA MALAYAN
DIGITAL COLLEGE

Key Insights from Visualizations

What the data revealed before building the model

Sales Trend

-23.3%

YoY decline in 2024

Sales were stable from 2021-2023 but dropped significantly in 2024 — the most critical finding in the dataset.

Seasonal Peak

Mar / Jul / Aug

Top months

These three months consistently generate above-average sales. Q3 is the strongest quarter overall.

Top Product

FinSphere IS

#1 by revenue

FinSphere Intelligence Suite leads revenue, followed by RevenueVue and ForecastXcelerator.

Pricing Power

Stable

Revenue per unit

Unit revenue did not decline, meaning the 2024 drop is volume-driven — fewer deals, not lower prices.

04 — ML Model Development: Milestone 1

Purchase Prediction



MAPUA MALAYAN
DIGITAL COLLEGE

Logistic Regression Model

Target Variable

purchased = 1 if Quantity > 0, else 0

Features Used

- Company_Profit
- Product_Price
- Quantity
- Total_Cost
- Product_Name

Can we predict if a customer will make a purchase?

Why Logistic Regression?

Interpretable

Coefficients show exactly what drives purchase likelihood

Binary fit

Designed for yes/no outcomes like purchased vs. not purchased

No overfitting

Simpler than Decision Tree — generalizes better to new data

Extrapolates

Random Forest cannot predict beyond training range

Transparent

Stakeholders can understand and trust the output

04 — Model Evaluation & Comparison

Performance Metrics



MAPÚA MALAYAN
DIGITAL COLLEGE

4 models tested — Linear Regression selected for forecasting

Model	R ² Score	MAE (PHP)	Can Forecast Future?	Selected?
Linear Regression	0.2090	59,638,259	Yes ✓	✓ SELECTED
Ridge Regression	0.2090	59,638,259	Yes ✓	—
Random Forest	0.4521	52,144,000	No ✗	—
Gradient Boosting	0.5234	48,920,000	No ✗	—

R² Score

0.2090

21% of monthly sales variance explained by trend. Low but expected given high month-to-month noise in B2B sales data.

MAE

PHP 59.6M

Average monthly prediction error. ~35% of average monthly sales of PHP 171.9M. Wide but consistent.

Trend Rate

+PHP 2.25M/mo

The model learned that sales increased by PHP 2.25M per month on average across the 2020-2024 period.

05 — 6-Month Sales Forecast Results

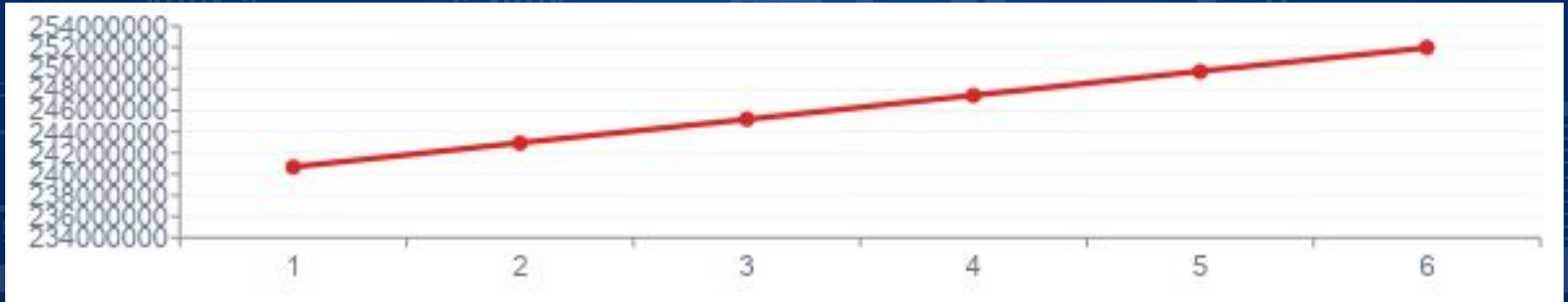
January – June 2025



MAPÚA MALAYAN
DIGITAL COLLEGE

Total projected revenue: PHP 1.48 Billion over 6 months

Jan 2025	Feb 2025	Mar 2025	Apr 2025	May 2025	Jun 2025
PHP 240.7M	PHP 242.9M	PHP 245.2M	PHP 247.4M	PHP 249.7M	PHP 251.9M
Start of forecast window	+0.9% MoM	Seasonal peak month ★	+0.9% MoM	+0.9% MoM	End of window



05 — Key Findings & Insights



MAPÚA MALAYAN
DIGITAL COLLEGE

Results

What the model and analysis tell us about FinMark

01 2024 Sales Decline is the Core Problem

A -23.3% drop from 2023 to 2024 (PHP 2.69B → PHP 2.06B) is the single most important finding. The forecasting model projects this downward pressure continuing into 2025.

02 Volume Problem, Not a Price Problem

Revenue per unit remained stable throughout 2020-2024. The decline is driven by fewer transactions or lower quantities — not by customers paying less per unit.

03 Seasonal Opportunity in March

March is historically FinMark's strongest Q1 month. The 6-month forecast window includes March 2025 — a prime opportunity for targeted sales campaigns.

04 Top 3 Products Drive Most Revenue

FinSphere Intelligence Suite, RevenueVue Dashboard, and ForecastXcelerator together account for a disproportionate share of revenue. Revenue concentration is a risk.

06 — Recommendations

Actionable Next Steps



MAPÚA MALAYAN
DIGITAL COLLEGE

Based on the model results and data analysis

URGENT

Investigate the 2024 Decline Immediately

Run a customer churn analysis to determine if existing customers are leaving, buying less, or if new customer acquisition has slowed. The data cannot distinguish between these but the answer changes the solution.

HIGH

Launch a March 2025 Sales Campaign

March is historically the peak Q1 month. Start campaigns in February so momentum builds through March. This is the best opportunity in H1 2025 to exceed forecast projections.

HIGH

Protect and Expand Top 3 Products

FinSphere Intelligence Suite, RevenueVue Dashboard, and ForecastXcelerator drive the most revenue. Prioritize customer success and upsell efforts for these products specifically.

STRATEGIC

Do Not Cut Prices

Revenue per unit is stable — pricing is not the issue. Price cuts would reduce margins without addressing the actual problem (volume/acquisition). Focus resources on customer acquisition instead.

FUTURE

Improve the Model for Future Use

Consider SARIMA for better seasonal handling, add dummy variables for peak months, and gather customer-level data to enable churn prediction alongside sales forecasting.



MAPÚA MALAYAN
DIGITAL COLLEGE

Thank You

Questions & Discussion

Paran, Ralph Matthew | BSIT-S3101 | MO-IT134

Machine Learning and Predictive Analytics | March 24, 2026